

AI Lead Management & Sales Automation

Agent Blueprint & Implementation Guide

Capture → Qualify → Prioritize → Automate

A reusable blueprint for understanding, deploying, testing, and extending the complete n8n workflow.

Document Purpose

This document explains the architecture, node responsibilities, data contracts, AI responsibilities, deterministic business logic, storage model, notification routing, customer communication, testing strategy, and deployment considerations.

Core Design Principle

AI understands and recommends. Deterministic workflow logic executes.

1. System Overview

The system takes an incoming customer enquiry through a structured lead-management lifecycle. Specialized AI agents handle language-heavy reasoning, while n8n nodes and JavaScript handle storage, routing, CRM operations, and communication.

```

Lead
↓
AI Qualification
↓
AI Sales Intelligence
↓
CRM Automation
↓
Responses + CRM Storage
↓
Notification Decision
■■■ TRUE → Internal Sales Notification
■■■ FALSE → Skip Internal Notification
↓
Customer Communication
↓
Respond to Webhook

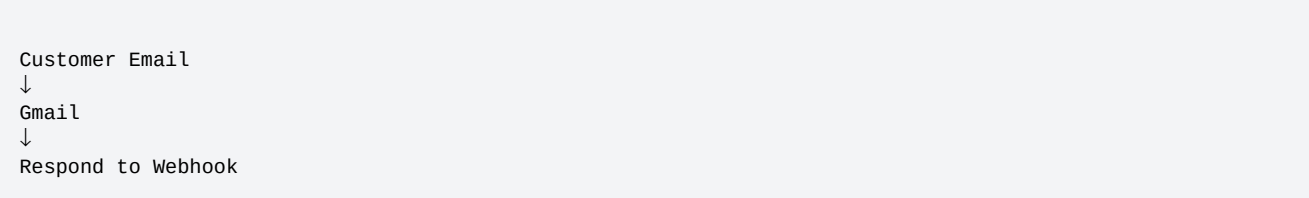
```

Primary Objectives

- Capture incoming leads reliably.
- Convert unstructured messages into structured lead intelligence.
- Evaluate sales value and follow-up priority.
- Persist analytical and operational data.
- Notify internal teams only when required.
- Send personalized customer communication without exposing internal sales data.
- Return a final workflow response through the webhook.

2. High-Level Architecture

```
Postman / Client  
↓  
Webhook  
↓  
Lead Capture  
↓  
AI Qualification Agent  
↓  
AI Sales Intelligence Agent  
↓  
CRM Automation Code  
↓  
██████████████████████████████████████████████████████████  
↓ ↓  
Responses Sheet CRM Sheet  
██████████████████████████████████████████████████████████  
↓  
Merge  
↓  
IF: Notification Required?  
/ \  
TRUE FALSE  
↓ ↓  
Internal Sales Skip  
Notification  
\ /  
\ /  
↓ ↓
```



3. Component & Node Blueprint

Component	Responsibility	Type
Webhook	Receives the lead request and returns the workflow result.	n8n
Lead Capture / Preparation	Preserves customer fields and prepares data for AI processing.	n8n / Code
AI Qualification Agent	Extracts lead summary, intent, category, pain point, urgency, sentiment, confidence, and qualification status, and next action.	AI Agent
Qualification Code	Parses and normalizes qualification JSON.	JavaScript
Responses Sheet	Stores lead and qualification/sales intelligence data.	Google Sheets
AI Sales Intelligence Agent	Produces score, temperature, priority, probability, response time, opportunity, recommendation, and follow-up requirements.	AI Agent
Sales Intelligence Code	Parses AI output and combines it with qualification data.	JavaScript
CRM Automation	Creates operational CRM fields and deterministic follow-up/notification logic.	JavaScript
CRM Sheet	Stores operational CRM information separately.	Google Sheets
Merge	Combines response and CRM records.	n8n
IF: Notification Required	Routes attention-worthy leads to internal notification.	n8n IF
Internal Gmail	Notifies the sales team with actionable lead and CRM details.	Gmail
Customer Email AI	Generates the customer-facing response.	AI Agent
Customer Gmail	Sends the customer email.	Gmail
Respond to Webhook	Returns the final workflow result.	n8n

4. Four-Stage Workflow

Stage 1 — Lead Capture

The workflow receives a customer name, email, and message through a webhook. The original lead information is preserved throughout the workflow.

```
{
  "name": "Chandu",
  "email": "example@email.com",
  "message": "Hello, I need help with a project."
}
```

Flow: Webhook → Lead Data → Initial Response → Storage → Customer Email

Stage 2 — AI Lead Qualification

The first analytical AI agent turns natural language into structured lead intelligence.

- Lead Summary
- Intent
- Category
- Pain Point

- Urgency
- Sentiment
- Confidence
- Qualification Status
- Next Action

```
{
  "lead_summary": "The customer is seeking personalized career guidance.",
  "intent": "Career Consultation",
  "category": "Consultation",
  "pain_point": "Needs career guidance",
  "urgency": "Medium",
  "sentiment": "Neutral",
  "confidence": 90,
  "qualification_status": "Qualified",
  "next_action": "Schedule Consultation"
}
```

Stage 3 — AI Sales Intelligence

A second AI agent evaluates the qualified lead from a business and sales perspective.

- Lead Score
- Lead Temperature
- Sales Priority
- Conversion Probability
- Response Time
- Opportunity Level
- Sales Recommendation
- Follow-up Required

```
{
  "lead_score": 97,
  "lead_temperature": "Hot",
  "sales_priority": "High",
  "conversion_probability": "High",
  "response_time": "Immediately",
  "opportunity_level": "High",
  "sales_recommendation": "Assign Sales Representative",
  "follow_up_required": true
}
```

Stage 4 — CRM & Follow-up Automation

The final stage converts sales intelligence into deterministic operational actions. JavaScript creates CRM state, ownership, follow-up date, internal notes, and the notification flag.

```
{
  "crm_status": "Pending Contact",
  "assigned_to": "Sales Team",
  "follow_up_date": "2026-08-06",
  "last_action": "CRM Record Created",
  "notification_required": true,
  "status": "CRM Updated"
}
```

5. Data Contracts

Incoming Lead

```
{
  "name": "Customer Name",
  "email": "customer@example.com",
  "message": "Customer enquiry"
}
```

Qualification Contract

```
{
  "lead_summary": "string",
  "intent": "string",
  "category": "string",
  "pain_point": "string",
  "urgency": "High | Medium | Low | Unknown",
  "sentiment": "Positive | Neutral | Negative",
  "confidence": 0,
  "qualification_status": "Qualified | Needs More Information | Unqualified",
  "next_action": "string"
}
```

Sales Intelligence Contract

```
{
  "lead_score": 0,
  "lead_temperature": "Hot | Warm | Cold",
  "sales_priority": "High | Medium | Low",
  "conversion_probability": "High | Medium | Low",
  "response_time": "Immediately | Within 2 Hours | Within 24 Hours | Within This Week",
  "opportunity_level": "High | Medium | Low",
  "sales_recommendation": "string",
  "follow_up_required": true
}
```

CRM Contract

```
{
  "crm_status": "string",
  "assigned_to": "string",
  "follow_up_date": "YYYY-MM-DD",
  "last_action": "string",
  "internal_notes": "string",
  "notification_required": true,
  "crm_updated_at": "ISO timestamp"
}
```

6. AI Agent Responsibilities

Qualification Agent

- Understand the customer's request.
- Identify intent and category.
- Summarize the requirement.
- Extract the main pain point.
- Estimate urgency and sentiment.
- Return confidence.
- Determine qualification status and next action.

Sales Intelligence Agent

- Use only supplied lead information.
- Calculate a consistent lead score.
- Map score to temperature, priority, probability, opportunity, and response time.

- Generate a sales recommendation.
- Determine whether follow-up is required.
- Return strict JSON for downstream automation.

Customer Communication Agent

- Personalize using qualification information.
- Acknowledge the enquiry.
- Guide the customer toward the recommended next step.
- Avoid exposing internal CRM/sales intelligence.
- Avoid inventing services, links, contact information, or commitments.

7. Deterministic Business Logic

Business rules should be implemented with code where predictable behavior is required.

Notification Rule

```
High Priority → notification_required = true
Medium Priority → notification_required = true
Low Priority → notification_required = false
```

IF Node

```
{{ $json["Notification Required"] }}
```

Condition:
is true

This keeps the notification decision predictable and easy to test.

8. Data Storage Blueprint

Responses Sheet

Group	Example Fields
Identity	response_id, lead_id, name, email
Original Request	message
Qualification	Lead Summary, Intent, Category, Pain Point, Urgency, Sentiment, Confidence
Qualification Decision	Qualification Status, Next Action
Sales Intelligence	Lead Score, Lead Temperature, Sales Priority, Conversion Probability, Response Time, Opportunity Level, Sales R
Follow-up	Follow up Required, Updated At

CRM Sheet

Field	Purpose
CRM Status	Current CRM state
Assigned To	Sales ownership/queue
Follow-up Date	Planned follow-up date
Last Action	Latest CRM operation

Field	Purpose
Internal Notes	Human-readable sales context
Notification Required	Controls internal notification
CRM Updated At	CRM update timestamp

9. Communication Blueprint

Internal Sales Email

The internal notification is for the sales/operations team and can include lead score, priority, CRM status, assigned team, follow-up date, recommendation, and internal notes.

Subject:
 ■ High Priority Lead - {{name}}

Body:
 Customer details
 Lead information
 Sales intelligence
 CRM information
 Recommended action
 Internal notes

Customer Email

The customer-facing email is intentionally separate from internal CRM information.

- Include the customer's name and relevant enquiry context.
- Acknowledge the request.
- Use the recommended next action when appropriate.
- Do not expose lead score, sales priority, CRM status, assigned team, internal notes, or notification flags.
- Do not invent booking links, services, phone numbers, or company information.

10. Testing Blueprint

Test Case A — High-Intent Project

Input:
 We want to build a custom AI automation platform.

Verify:

- Qualification is produced.
- Sales intelligence is produced.
- CRM record is created.
- Notification Required becomes true when business rules classify it as high/medium attention.
- Internal notification is sent.
- Customer email is sent.
- Webhook completes.

Test Case B — Low-Intent Enquiry

Input:
 Hi, I'm just browsing your services and gathering some general information.
 I don't have any immediate requirements.

Verify:

- The lead is not incorrectly forced into a high-priority path.
- Notification Required becomes false when business rules classify it as low priority.
- Internal notification is skipped.
- Customer communication still completes.

Test Case C — Needs More Information

Input:

A vague enquiry with insufficient project details.

Verify:

- Qualification identifies missing information.
- Customer communication requests necessary details.
- Sales recommendation remains consistent with qualification.
- No unsupported assumptions are introduced.

11. Reliability & Error Handling

- Validate AI JSON before using it downstream.
- Use deterministic defaults when parsing fails.
- Preserve original lead data through every stage.
- Do not let missing optional fields break customer communication.
- Keep CRM data separate when the main sheet schema should remain stable.
- Test both TRUE and FALSE notification paths.
- Keep credentials outside workflow exports where possible.
- Log timestamps for important state changes.

12. Security Considerations

- Never publish API keys, OAuth tokens, passwords, or private webhook URLs.
- Use n8n credential management instead of hard-coding secrets.
- Avoid exposing internal CRM information in customer-facing messages.
- Restrict access to CRM and sales notification destinations.
- Validate external input before using it downstream.
- Move to a production database/CRM when scale, access control, or concurrency requirements exceed Google Sheets.

13. Setup & Deployment

Prerequisites

- Running n8n instance.
- Configured AI model/provider.
- Google Sheets credentials.
- Gmail credentials.
- Postman or another HTTP client.

Basic Setup

Import the workflow into n8n.

Configure AI credentials.

Configure Google Sheets credentials.

Configure Gmail credentials.

Connect the Responses Sheet and CRM Sheet.

Verify AI prompts and output contracts.

Configure the webhook.

Run high- and low-priority tests.

Verify Google Sheets and email outputs.

Activate the workflow.

Security: Never commit API keys, OAuth tokens, passwords, private webhook URLs, or other credentials.

14. Future Extension Blueprint

- Replace Google Sheets with PostgreSQL or Supabase.
- Integrate HubSpot, Salesforce, or another dedicated CRM.
- Add Slack or Discord notifications.
- Build a sales dashboard.
- Add persistent follow-up reminders.
- Add retry and error-handling workflows.
- Add human approval for high-impact actions.
- Add authentication and role-based access.
- Add conversion analytics.
- Introduce queue-based processing for larger workloads.

15. Final Workflow Summary

```
Customer Enquiry
↓
Lead Capture
↓
AI Qualification
↓
AI Sales Intelligence
↓
CRM Automation
↓
Responses + CRM Storage
↓
Merge
↓
Notification Decision
■ ■
TRUE FALSE
↓ ↓
Sales Alert Skip Alert
■ ■
Customer Email
↓
Gmail Delivery
↓
Webhook Response
```

Core lifecycle: Capture → Qualify → Prioritize → Automate

The blueprint is designed to be adapted to different business domains while maintaining separation between AI reasoning, deterministic workflow execution, internal operations, and customer communication.